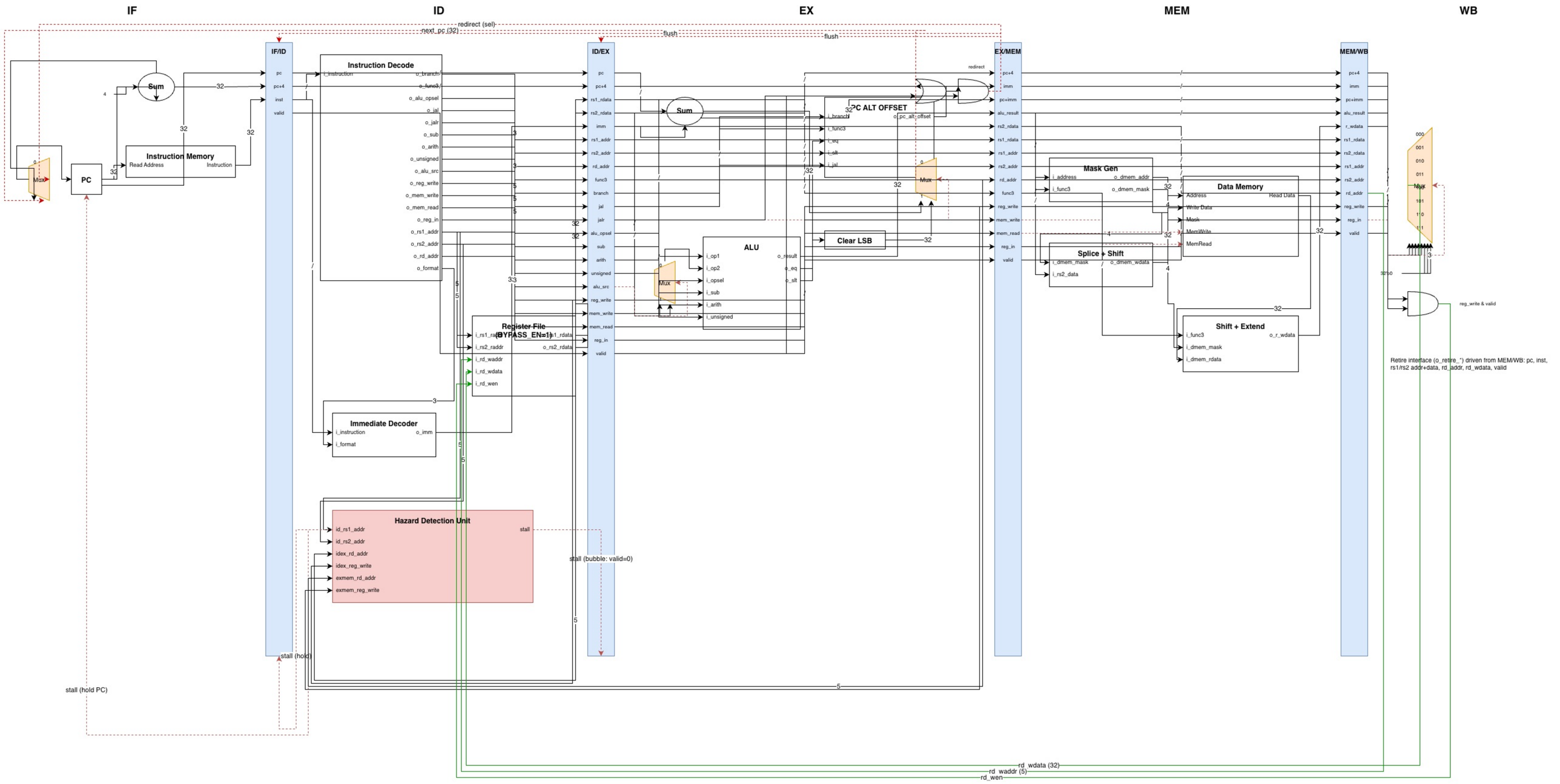

IF
ID
EX
MEM
WB
redirect (sel)
next_pc (32)
flush
flush
IF/ID
ID/EX
EX/MEM
MEM/WB
Sum
PC
Instruction Memory
Read Address
Instruction
Instruction Decode
i_instruction
o_branch
o_func3
o_alu_opsel
o_jal
o_jalr
o_sub
o_arith
o_unsigned
o_alu_src
o_reg_write
o_mem_write
o_mem_read
o_reg_in
o_rs1_addr
o_rs2_addr
o_rd_addr
o_format
Register File
(BYPASS_EN=1)
i_rs1_raddr
i_rs2_raddr
i_rd_waddr
i_rd_wdata
i_rd_wen
o_rs1_rdata
o_rs2_rdata
Immediate Decoder
i_instruction
i_format
o_imm
Hazard Detection Unit
id_rs1_addr
id_rs2_addr
idex_rd_addr
idex_reg_write
exmem_rd_addr
exmem_reg_write
stall
stall (bubble: valid=0)
stall (hold)
stall (hold PC)
Sum
PC ALT OFFSET
i_branch
i_func3
i_eq
i_slt
i_jal
o_pc_alt_offset
Clear LSB
ALU
i_op1
i_op2
i_opsel
i_sub
i_arith
i_unsigned
o_result
o_eq
o_slt
Mux
redirect
Mask Gen
i_address
i_func3
o_dmem_addr
o_dmem_mask
Splice + Shift
i_dmem_mask
i_rs2_data
o_dmem_wdata
Data Memory
Address
Write Data
Mask
MemWrite
MemRead
Head Data
Shift + Extend
i_func3
i_dmem_mask
i_dmem_rdata
o_r_wdata
reg_write & valid
Retire interface (o_retire_*) driven from MEM/WB: pc, inst, rs1/rs2 addr+data, rd_addr, rd_wdata, valid
rd_wdata (32)
rd_waddr (5)
rd_wen
Legend: solid = data, dashed red/brown = control, red = redirect/flush from EX, green = writeback to RF. Widths labelled at first appearance; '/' = multi-bit bus already labelled.